EE2026: DIGITAL DESIGN
Academic Year 2025-2026, Semester 2

FPGA Design Project (FDP)

GROUPING

Students are required to form their own team of 4 and register during the lab session in week 7. Depending on class size, we may need to create one to three teams of 3. We will arrange for all students without a team to meet each other during the lab session in week 7. Then you can form your own team after checking with each other.

Teams that consist of students who have not already submitted ALL THREE assignments of the fundamental labs will not be automatically approved. Contact the lab instructors.

ABSTRACT

A FPGA Design Project (FDP) that makes use of the Basys3 development board and the following devices will need to be created:

Device	Description
Computer Mouse *(Optional)* *1 piece / team*	Clicks and movements on this device will provide digital signals to the FPGA
Pmod OLED *1 piece / member*	Information can be displayed on this 96x64 pixels display screen with 16-bit colour resolution

Deadline extensions are not given for damaged components. Use them with upmost care!

This FPGA project requires student to apply their fundamental programming skills in higher-order thinking tasks. The week 7 lab session **briefly guides** you in getting started with a basic system consisting of the OLED. Exact steps are not provided, and you are expected to code, observe, and make conclusions!

After some basic tasks have been implemented, you will need to use your own technical skills, logic, creativity, and resourcefulness to create an Interactive FDP (I-FDP), while deciding what is best **implemented within the resource limitations of the FPGA**.

PLAGIARISM WARNING

This is a teamwork project. Discussions are encouraged. However, 'discussion' is not a valid excuse if your codes turn out to be uncomfortably similar. NUS and the EE2026 teaching team take plagiarism very seriously.

- Each individual carries responsibility for **both** the team and individual portions of the project. Claiming for work done by, and not limited to, senior students, family members, relatives, paid or unpaid service providers, etc. are serious offences.
- All members of the team of the project solutions (codes) or reports will be **unconditionally penalised** in such cases.
- Warning from the NUS Code of Student Conduct:
  "Any student found to have committed or aided and abetted the offence of plagiarism may be subject to disciplinary action" https://www.nus.edu.sg/celc/programmes/plagiarism.html
- Students will also be reported to BoD (Board of Discipline).

1. SCHEDULE AND PROJECT COMPONENTS

The project components of EE2026 makes up 30% of the course grade.

Week	Suggested Task for Each Week	Marks	Evaluation
7	**INDIVIDUAL BASIC TASK (4.P, 4.Q, 4.R, 4.S, 4.T)**		All the basic tasks of the FDP are evaluated at the start of the **week 8 lab session**. CANVAS **submission is required** before your official week 8 lab session. (See section 4.T)
	Student P: Basic task 4.P (All Parts of 4.P)	1.5% (Individual)	
	Student Q: Basic task 4.Q (All Parts of 4.Q)	1.5% (Individual)	
	Student R: Basic task 4.R (All Parts of 4.R)	1.5% (Individual)	
	Student S: Basic task 4.S (All Parts of 4.S)	1.5% (Individual)	
7, 8, 9, 10, 11, 12 **(Some weeks do not have labs)**	**I-FDP PROPOSAL DOCUMENT**		CANVAS **submission is required** before the end of week 8. (See Section 6)
	Group: I-FDP proposal - Indicate group theme, task responsibilities and timeline	1.5% (Group)	
	I-FDP IMPLEMENTATION		I-FDP is evaluated in week 12, 13 in **ONE** bitstream for the team. Multiple bitstreams are **not** allowed. CANVAS **submission is required** in week 12. (See Section 7) The basic tasks that have already been assessed in week 8 are not required in the bitstream and are not re-assessed.
	Student A: Component A of I-FDP	9% (Individual)	
	Student B: Component B of I-FDP	9% (Individual)	
	Student C: Component C of I-FDP	9% (Individual)	
	Student D: Component D of I-FDP	9% (Individual)	
	Group: Full integration of I-FDP components A, B, C and D in 1 bitstream	9% (Group)	
12, 13	**OVERALL EXECUTION, Q&A, UNDERSTANDING, COMPLETION OF PEER FEEDBACK**	9% (Individual)	Evaluated during presentation, through report, and from peer feedback

2. SOFTWARE FILES

A project template (**FDP.xpr.zip**) can be downloaded from CANVAS. The template consists of the following design sources:

Design Sources:

Top_Student.v: The top-module of the design. This will be your main module, or typically called the Top Level module, where you instantiate the sub-modules and make the necessary links between these sub modules. **You will need to modify this module, as well as create other design sources for use in this module.**

Oled_Display.v: An interface module between your design on the FPGA and the OLED display. This module works to send serial SPI data into the OLED display for initialization and drawing. **You are NOT required to make changes to this module. Optionally, you should read the description inside if you want to know more about the input and output signals.**

MouseCtl.vhd: An interface module between the computer mouse, Ps2Interface.vhd and your design on the FPGA. This module signals when clicks and movements are detected on the computer mouse connected to the Basys3 USB port. A wired mouse is recommended, as not all wireless mice will work. **You are NOT required to make changes to this module. This module is optional, and its usage will not be explained.**

Paint.v: A module that connects the computer mouse and OLED, so that digits created by the mouse on the OLED can be recognised, and the results shown on the seven segments display and LEDs on the Basys3. **You are NOT required to make changes to this module. This module is optional, and its usage will not be explained.**

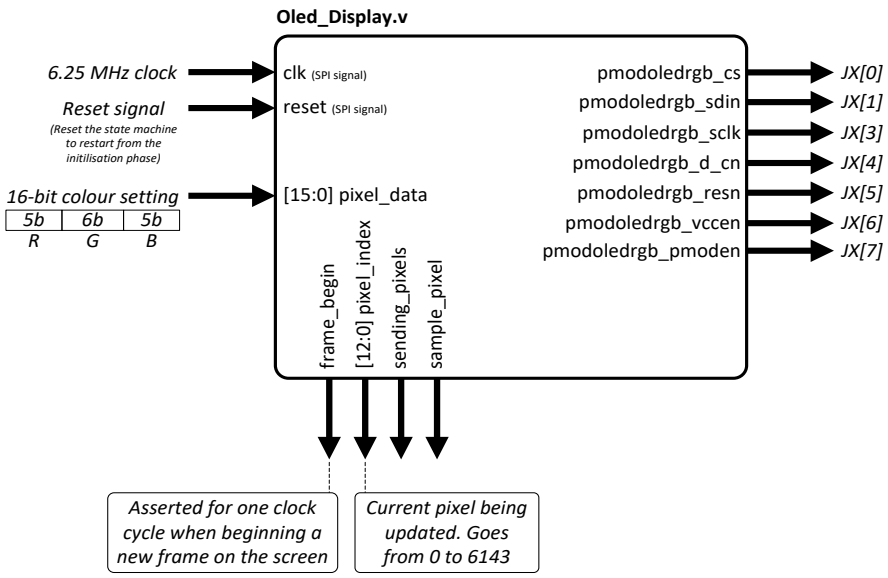
Constraint Sources:

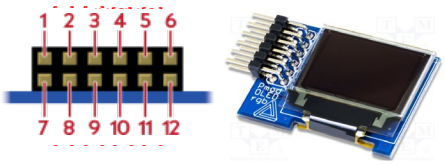
Basys3_Master.xdc: A master constraints file that defines the I/O constraints for the Basys 3 Development Board

3. MODULE SETUP AND TESTING GUIDELINES

3.A OLED_DISPLAY.V MODULE

Setting up the OLED by using `Oled_Display.v`





3.A1 **Instantiate the** `Oled_Display.v` **module in** `Top_Student.v`

3.A2 **Create and provide a 6.25 MHz clock signal named** `clk6p25m`**, to** `Oled_Display.v`

3.A3 **Create a 16-bit signal named** `oled_data` **and initialize it with a value of** `16'h07E0`.
The `oled_data` defines the colour of each pixel. The RGB OLED screen has 16-bit colour resolution, this is represented through 5 bits for the red colour component, 6 bits for the green colour component, and 5 bits for the blue colour component. Connect `oled_data` to the `pixel_data` input of `Oled_Display.v`

3.A4 **Connect the** `Jx[7:0]` **signals to the PmodOLED device accordingly**.

3.A5 **Attach the OLED display to the Basys3 and update the constraints file accordingly**.
Very important: To avoid damaging the devices, ensure the Basys3 board is OFF before connecting the OLED
DO NOT touch the OLED ribbon cable when inserting / removing the OLED from the Basys3 board.

3.A6 **Optionally connect the** `reset` **port of** `Oled_Display.v` **to any pushbutton, or alternatively give the** `reset` **port a constant value of 0**.

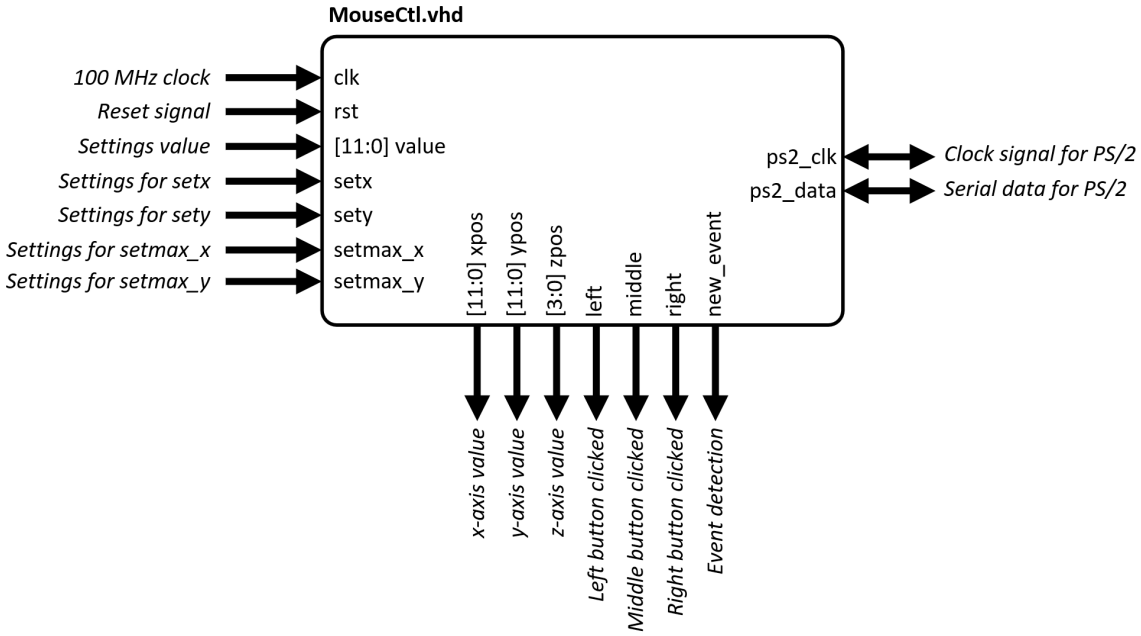
3.A7 **Generate the bitstream and download to the FPGA.**
Verify that the background colour of the screen is green.

3.A8 **Write the codes that uses sw[4] to select between one of these two possibilities:**
- sw[4] is OFF: The background colour is green.
- sw[4] is ON: The background colour is red.

3.A9 **Generate the bitstream and download to the FPGA**.
If the above steps are executed correctly, you would observe a green background on the OLED when sw[4] is OFF, and a red background when sw[4] is ON.

3.B MOUSECTL.VHD MODULE [This optional (And not explained) component may be used in section 5: I-FDP]

Setting up the computer mouse by using `MouseCtl.vhd`



3.B1 Instantiate the `MouseCtl.vhd` module in `Top_Student.v`

3.B2 Provide a 100 MHz clock signal to `MouseCtl.vhd`

3.B3 Connect ps2_clk port and ps2_data port.
`ps2_clk` is an inout signal which is connected to the FPGA PACKAGE_PIN C17.
`ps2_data` is an inout signal which is connected to the FPGA PACKAGE_PIN B17.

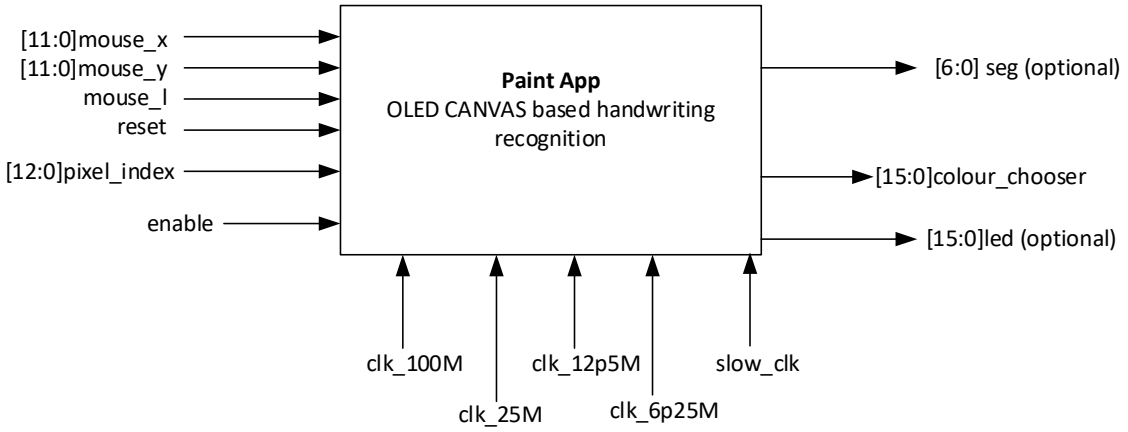
3.B4 Set default values for `value`, `setx`, `sety`, `setmax_x`, `setmax_x`.
There is no need to offset any of these values, and we can assume the origin to be the default.

3.B5 Write the code that turns on certain leds based on the mouse clicks.
led[15] must turn ON when the signal from "`left`" port is high.
led[14] must turn ON when the signal from "`middle`" port is high.
led[13] must turn ON when the signal from "`right`" port is high.

3.B6 Generate the bitstream and download to the FPGA.
Ensure that the mouse is connected to the (upper right) USB port of your Basys3 board. A wired mouse is recommended for testing purposes. Download your program to the FPGA. Press any of the three mouse buttons and observe which LED lights up on the Basys3 development board.

3.C PAINT.V MODULE [This optional (And not explained) component may be used in section 5: I-FDP]

Setting up the OLED based canvas handwritten recognition by using `Paint.v`



The Paint App module is a CANVAS drawing app with a built-in handwritten-digit recognition using the neural network algorithm. It contains 11 inputs, which are mouse_x, mouse_y, mouse_l, reset, pixel_index, clk_100M, clk_25M, clk_12p5M, clk_6p25M, slow_clk, and enable, and 3 outputs which are seg, colour_chooser and led. Students to take the following procedure to properly instantiate the Paint App.

3.C1 Instantiate the paint.v **module in** Top_Student.v

3.C2 Instantiate both the MouseCtl.vhd **and** Oled_Display.v **modules in** Top_Student.v

3.C3 Connect mouse_x_pos to mouse_x, mouse_y_pos to mouse_y, **and** mouse_left_click to mouse_l **between** MouseCtl.vhd **and** Paint.v **ports.**

3.C4 Assign mouse_right_click **to** reset **then connect it between between** MouseCtl.vhd **and** Paint.v **ports.**

3.C5 Connect Pixel Index **to** pixel_index**, and** colour_chooser **to** pixel_data **between** Oled_Display.v **and** Paint.v.

3.C6 Provide the clock signal with follow frequencies to Paint.v

- **100 MHz via** clk_100M
- **25 MHz via** clk_25M
- **12.5 MHz via** clk_12p5M
- **6.25 MHz via** clk_6p25M
- **Between 0.5 to 2.0 Hz to slow_**clk

3.C7 Connect LED **to** led **and** seven segment display **to** seg **to** Paint.v**.**

3.C8 Check the following files are available under "Coefficient Files": 00.coe **and** weights.coe**. If any of these files is unavailable, add these files via "**.\MODS.srcs\sources_1\new**" using add design sources wizards.**

3.C9 Click the "IP sources**" under** Source Tab **(where you browse the Verilog file). Check the following memory blocks are available:** blk_mem_gen_0, blk_mem_gen_const, blk_mem_gen_img **and** blk_mem_gen_inter**. If any of the memory block is missing, add the** *.dcp **memory block file via "**.\MODS.srcs\sources_1\ip**" using add design sources wizards, and right click to choose "**Generate Output Products**"**

3.C10 Instantiate both mouse and OLED, generate the bitstream and download to the FPGA
Ensure that the mouse is connected to the (upper right) USB port of your Basys3 board, and OLED is connected to corresponding pmod pin. A wired mouse is recommended for testing purposes. Download your program to the FPGA. Use mouse to choose the color on OLED and write any number (and only 1) by holding down "LEFT MOUSE BUTTON" on the white CANVAS. Observe the output from seven segment display and LED on basys 3 board. "RIGHT CLICK" to reset the CANVAS.

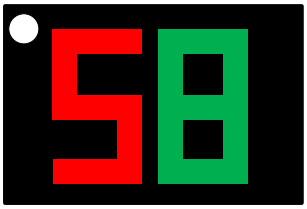
4. WEEK 7 BASIC TASKS – ASSESSED DURING WEEK 8 LAB SESSION

The basic tasks help students to get familiarised with the usage of the OLED screen. Several requirements are similar between the different basic tasks, so **team members can learn by sharing knowledge between themselves**!
If your team consists of 3 persons, ignore any one of the four individual basic tasks (4.P or 4.Q or 4.R or 4.S)

OLED COORDINATES SYSTEM:

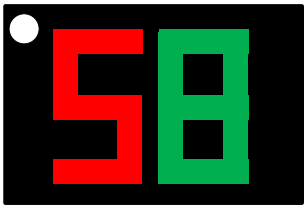
Work on creating an **x** and **y** coordinate system derived from `pixel_index` to make the pixel drawings on the OLED screen less cumbersome (*Hint: You may employ the modulus % and / operators*)

4.P STUDENT P: BASIC TASK P



Two characters and solid white circle

Clearing the OLED by pressing BTNU



Pressing BTNU to revert to the original screen

Consider the **2nd Rightmost Number**, **1st Rightmost Number** of your student number:

	7th Rightmost Number	6th Rightmost Number	5th Rightmost Number	4th Rightmost Number	3rd Rightmost Number	2nd Rightmost Number	1st Rightmost Number	Rightmost Alphabet
A	**0**							

4.P1 When the program starts, the following appears on the black OLED screen:

- The **2nd Rightmost Number** appears on the left side of the OLED in **red colour**
- The **1st Rightmost Number** appears on the right side of the OLED in **green colour**
- A solid **white circle**, with radius **between 4 to 8 pixels**, appears near the top left of the OLED screen
  Hint: Consider using the equation of a circle to draw the solid circle

Use the number template shown below as the requirement for a readable character for this basic task:

0 123456789	Height of each character: **At least** 45 pixels Width of each character: **At least** 25 pixels Lines thickness: **At least** 6 pixels thick

4.P2 If both characters and solid white circle are present on the OLED, the user can press the up pushbutton (BTNU) to clear them from the screen.

If the screen is empty, the user can press the up pushbutton (BTNU) to make the two characters and solid white circle appear on the screen again.

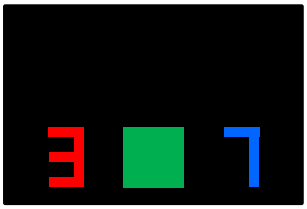
This process can happen an infinite number of times.

Very important note regarding the pushbutton presses:

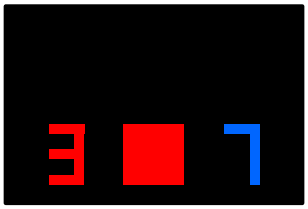
- The user may press and release the pushbutton within 200 milliseconds (No need to count the actual duration the user pressed the pushbutton, or if 200 milliseconds have been exceeded). The pushbutton signal detection must be **very responsive** to this quick pressing (Example: Detection done every 1 millisecond) and **must apply debouncing** to avoid multiple presses detection. For debouncing, create a counter that counts for 200 milliseconds at the start of the pushbutton pressing, and during that time, bouncing pushbutton signals must be ignored.

- The user may also press and hold for more than 200 milliseconds. In such case, the pushbutton press should be detected as being just one press, no matter how long the pushbutton is pressed and held. It may be held for as long as 5 seconds.

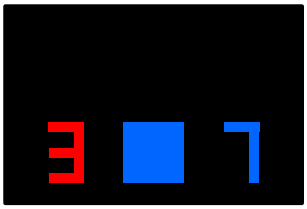
4.Q STUDENT Q: BASIC TASK Q



Two characters and a green solid square



Solid square change colour by pressing BTND



Solid square change colour by pressing BTND

Consider the **4th Rightmost Number, 2nd Rightmost Number** of your student number:

	7th Rightmost Number	6th Rightmost Number	5th Rightmost Number	4th Rightmost Number	3rd Rightmost Number	2nd Rightmost Number	1st Rightmost Number	Rightmost Alphabet
A	**0**							

4.Q1 When the program starts, the following appears on the black OLED screen:

- A **green 20x20 pixels solid square** on the bottom middle of the OLED
- A **red character representing the 4th rightmost number** on the bottom left of the OLED
- A **blue character representing the 2nd rightmost number** on the bottom right of the OLED

Use the number template shown below as the requirement for a readable character for this basic task:

0 123456789	Height of the character: **Exactly** 20 pixels Width of the character: **Maximum** of 20 pixels Lines thickness: **Between** 2 to 5 pixels thick

4.Q2 The solid square on the bottom middle of the OLED can change colour. Each time the down pushbutton (BTND) is pressed, the colour for the solid square changes in this sequence (one colour change per pushbutton press):

Green > Red > Blue > Yellow > "Back to Green" (Sequence repeats)

Very important note regarding the pushbutton presses:

- The user may press and release the pushbutton within 200 milliseconds (No need to count the actual duration the user pressed the pushbutton, or if 200 milliseconds have been exceeded). The pushbutton signal detection must be **very responsive** to this quick pressing (Example: Detection done every 1 millisecond) and **must apply debouncing** to avoid multiple presses detection. For debouncing, create a counter that counts for 200 milliseconds at the start of the pushbutton pressing, and during that time, bouncing pushbutton signals must be ignored.

- The user may also press and hold for more than 200 milliseconds. In such case, the pushbutton press should be detected as being just one press, no matter how long the pushbutton is pressed and held. It may be held for as long as 5 seconds.

4.R STUDENT R: BASIC TASK R

Two characters when the program starts

Blue character moves when switch SW1 is turned ON

Blue character stops when switch SW1 is turned OFF

Arrows are not present on the OLED. They are only for illustration purposes of the movement directions.

Consider the **3rd Rightmost Number**, **1st Rightmost Number** of your student number:

	7th Rightmost Number	6th Rightmost Number	5th Rightmost Number	4th Rightmost Number	3rd Rightmost Number	2nd Rightmost Number	1st Rightmost Number	Rightmost Alphabet
A	**0**							

4.R1 When the program starts, the following appears on the black OLED screen:

- The **1st Rightmost Number** appears on the left side of the OLED in **blue colour**
  To improve visibility of the number on the OLED, use a lighter blue colour!
- The **3rd Rightmost Number** appears in the middle of the OLED in **orange colour**

Use the number template shown below as the requirement for a readable character for this basic task:

	Height of each character: **Between** 20 to 30 pixels Width of each character: **Between** 15 to 20 pixels Lines thickness: **Exactly** 4 or 5 pixels thick

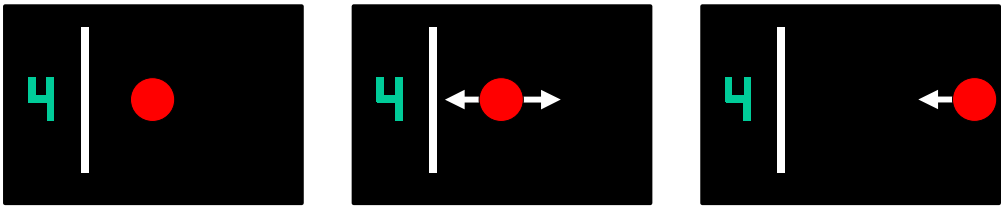
4.R2 If switch **SW1** is turned ON, the blue number moves between the right edge and left edge of the screen. The movement speed must be such that is takes around **3 ± 0.5 seconds** to move from one edge to the opposing edge. For ease of programming, there is no need to be exactly at the edge (It can be within 3 pixels from the edges)

If **SW1** is turned OFF, the movement of this **blue number** must stop, and if **SW1** is turned ON, it resumes in the same direction that it was going before it stopped. **SW1** can be turned ON / OFF an infinite number of times.

<u>Important note regarding the number movement:</u>

The movement animation should be smooth. It should not appear as if the moving number is skipping pixels or teleporting itself to move. There is also no restriction on which number is in the foreground, and which number in the background when they cross path near the middle of the screen.

4.S STUDENT S: BASIC TASK S



Character, wall and solid circle | *Solid circle can move left or right* | *Solid circle can move left when at the right edge*

Arrows are not present on the OLED. They are only for illustration purposes of the movement directions.

Consider the **2nd Rightmost Number** of your student number:

	7th Rightmost Number	6th Rightmost Number	5th Rightmost Number	4th Rightmost Number	3rd Rightmost Number	2nd Rightmost Number	1st Rightmost Number	Rightmost Alphabet
A	**0**							

4.S1 When the program starts, the following appears on the black OLED screen:

- A **solid red circle**, with radius **between 9 to 15 pixels**, appears near the centre of the OLED screen
  Hint: Consider using the equation of a circle to draw the solid circle
- A **white wall of 5x50 pixels** appears at around the left quarter of the OLED
- The **2nd Rightmost Number** appears between the OLED left edge and the white wall in **green colour**
  To improve visibility of the number on the OLED, use a brighter or lighter green colour!

Use the number template shown below as the requirement for a readable character for this basic task:

0 123456789	Height of each character: **Between** 12 to 16 pixels Width of each character: **Between** 6 to 9 pixels Lines thickness: **Exactly** 3 pixels thick

4.S2 The user can press (**Do not hold**) certain pushbuttons to make the **solid circle** move automatically (No need to keep pressing or holding the pushbutton) in a certain direction till near the OLED screen edge, or till it hits the wall. It stays stationary after hitting the wall or the edge of the OLED screen. The movement speed for the **solid circle** is approximately **35 to 55 pixels per second**.

The directions for the solid circle are as indicated below (Pushbuttons are pressed and released immediately):

- **Right pushbutton** for continuous /automatic **right** movement of the solid circle.
- **Left pushbutton** for continuous / automatic **left** movement of the solid circle.

The solid circle can change directions while it is moving (If moving right, and left pushbutton is pressed, the new direction will be left. If moving left, and right pushbutton is pressed, the new direction will be right.)

Note regarding the pushbutton presses:

The user will press and release the pushbutton within 100 milliseconds (No need to count the actual amount of time the user pressed the pushbutton, or if 100 milliseconds have been exceeded). The pushbutton signal detection must be **very responsive** to this quick pressing (Example: Detection done every 1 millisecond). Debouncing is optional here as it does not matter if a pushbutton press is detected as multiple presses.

Important note regarding the number movement:

The movement animation should be smooth. It should not appear as if the moving solid circle is skipping pixels or teleporting itself to move.

4.T GRADING OF THE BASIC TASKS IN WEEK 8

The basic tasks completed by the team members are required to be integrated into **ONE single bitstream**. Team members whose basic task are integrated within that **ONE single bitstream** can obtain up to the full marks of 1.5% assigned to the basic task component.

To emphasise on the importance of planning for integration, full marks will not be given if any of the following scenario occurs for a specific student (Other team members are not affected):

- Student's basic task was not integrated within that **ONE single bitstream** unanimously decided by the team
- Student's basic task was **shown later or shown separately** from the team's presentation during the lab session

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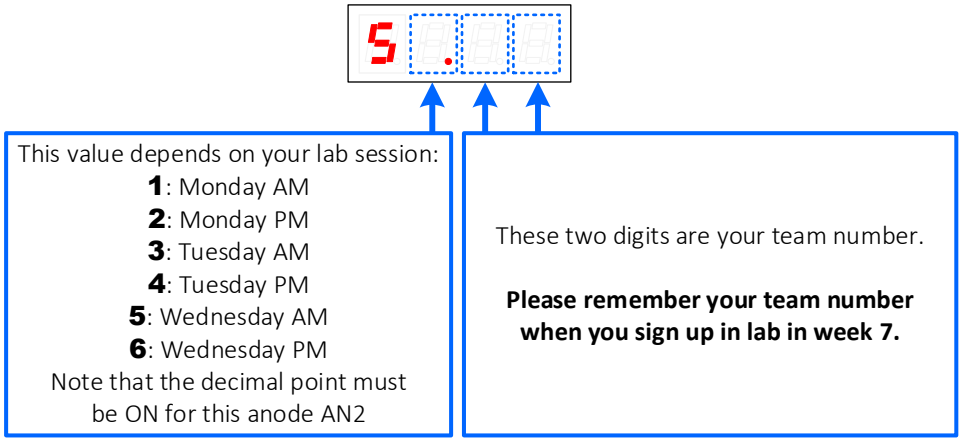
For the **ONE single bitstream**, when the program starts, the seven segments displays must show your group ID in this format (It is compulsory for <u>all students</u> to know how to get different characters to appear simultaneously on the seven segments displays):



The basic task to present should be based on the following truth table:

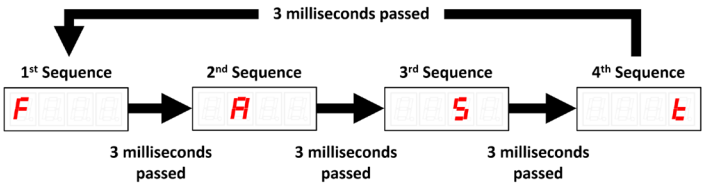
INPUTS				OUTPUTS
SW15	SW14	SW13	SW12	OLED
0	0	0	1	Basic Task P for Student P (If present in bitstream)
0	0	1	X	Basic Task Q for Student Q (If present in bitstream)
0	1	X	X	Basic Task R for Student R (If present in bitstream)
1	X	X	X	Basic Task S for Student S (If present in bitstream)

You will be **allowed to re-program** the same ONE single bitstream to the FPGA to show each student's basic tasks from beginning.

Hint: To be able to show different characters on the seven-segments displays, it is required to:

- *Show a character on only 1 anode at a time*
- *Show another character on another anode a few milliseconds later (around 200 to 500 Hz)*
- *Repeat the above continuously*

An example of the above task is illustrated below:



For humans, this is what they will see:

UPLOAD DETAILS FOR WEEK 7 BASIC TASKS:

- One **single archive** (.xpr.zip) **per group**

- File name format: **Week_7_SX_YZ_Group**, where **SX_YZ** is your group ID
  (See PDF on CANVAS near submission time for your **SX_YZ**).

- Upload in the CANVAS folder named: **Week 7 FDP Submission**

- Deadline for submission:
  - **Monday AM/PM Sessions:** Upload to be completed by **Monday 9th March 2026, 6 AM (Morning)**
  - **Tuesday AM/PM Sessions:** Upload to be completed by **Tuesday 10th March 2026, 6 AM (Morning)**
  - **Wednesday AM/PM Sessions:** Upload to be completed by **Wednesday 11th March 2026, 6 AM (Morning)**

- The team must **submit one hardcopy** of the **completed** "Personalised Basic Tasks Form" during the lab session in week 8. The form is found on the next page of this lab manual.

- Late assessments of week 7 basic tasks are not accepted after your lab session in week 9

Submit one hardcopy of the completed "Personalised Basic Tasks Form" during the lab session in week 8:

PERSONALISED BASIC TASKS FORM:

Each student has a personalised basic task requirement based on the student matriculation number. Carefully write down your number, as mistakes in the student matriculation number used are not accepted.

Team ID (Format is SX_YZ. For Example, S8_03) ➔	

BASIC TASK P	Name of <Student P> as written on the Student Matriculation Card ➔	

	7th Rightmost Number	6th Rightmost Number	5th Rightmost Number	4th Rightmost Number	3rd Rightmost Number	2nd Rightmost Number	1st Rightmost Number	Rightmost Alphabet
A	0							

BASIC TASK Q	Name of <Student Q> as written on the Student Matriculation Card ➔	

	7th Rightmost Number	6th Rightmost Number	5th Rightmost Number	4th Rightmost Number	3rd Rightmost Number	2nd Rightmost Number	1st Rightmost Number	Rightmost Alphabet
A	0							

BASIC TASK R	Name of <Student R> as written on the Student Matriculation Card ➔	

	7th Rightmost Number	6th Rightmost Number	5th Rightmost Number	4th Rightmost Number	3rd Rightmost Number	2nd Rightmost Number	1st Rightmost Number	Rightmost Alphabet
A	0							

BASIC TASK S	Name of <Student S> as written on the Student Matriculation Card ➔	

	7th Rightmost Number	6th Rightmost Number	5th Rightmost Number	4th Rightmost Number	3rd Rightmost Number	2nd Rightmost Number	1st Rightmost Number	Rightmost Alphabet
A	0							

FOR GRADER'S USE:

I confirm that the student matriculation numbers match the official NUS records.

(Initials of grader: _____)

5. WEEK 7 TO WEEK 12: THE INTERACTIVE FDP (I-FDP)

An Interactive FPGA Design Project (I-FDP) is required to be implemented **within the resource limitations of the FPGA** on the Basys3 board, in **ONE** bitstream. Working together as a group, divide the responsibilities among the team members to achieve the following specifications for the I-FDP:

Group Theme	FPGA Based Smart Graphical Calculator using Verilog Language
Functionality	**Compulsory requirement:** Able to perform simple additions and subtractions of whole numbers (with single digits for operands, such as 8 + 5). **Possible enhancements (non-exhaustive):** Multiplication, division, decimal points, different number representations, trigonometry, graphing, conversions, educational components etc.
Robustness and Reliability	**Compulsory requirement:** Reprogramming the FPGA is not required for each calculation. System does not crash often due to unexpected behaviours. **Possible enhancements (non-exhaustive):** Switching between different functionalities always work, reset in-built, reliable communication between multiple Basys3 board if there are multiple boards etc.
User Interface	**Compulsory requirement:** Usage of switches, LEDs and OLED. **Possible enhancements (non-exhaustive):** Usage of a seven-segments display, multiple OLEDs, mouse, computer keyboard etc. *Note: Usage of unnecessary devices is discouraged (It negatively affects the product)*
User Experience	**Compulsory requirement:** User can input values, and see output values, on the Basys3 board, with ease. **Possible enhancements (non-exhaustive):** Intuitive usage of the calculator with minimal documentation, adequate on-board guidance where required, brightness/mode of the OLED, elements of fun and excitement when using the product, etc.

Compulsory requirements are the bare minimum required to show your fundamental Verilog capabilities.

Possible enhancements are value-added components that further demonstrate your Verilog capabilities and are not as straightforward to efficiently implement. The enhancement(s) must be strongly related to the group theme. Each team is encouraged to work together on a larger enhancement by appropriately dividing the responsibilities, instead of multiple unrelated / simpler enhancements, for the I-FDP.

When working on possible enhancements, note that it is not compulsory to have all the basic capabilities of a calculator. For example, some teams may not wish to implement inefficiently coded multiplication/division and instead focus on interactive control of graphs due to FPGA resource limitations.

Each team member must have a reasonable amount of Verilog coding to do, and their allocated responsibility should not have much overlap with the other members of the team. These individual responsibilities would form the marks for the individual marks of the I-FDP implementation.

The group marks of the I-FDP implementation consider how well the different components from each team member are integrated together into one bitstream to create a more complete project. It must not be a separate component.

ALTERNATIVE GROUP THEME OPTION FOR I-FDP:

To allow teams to further explore other options that they wish to implement using the Basys3 and Verilog as the building block, teams are encouraged to choose other group **themes that match their interests**, instead of the FPGA Based Smart Graphical Calculator described above. In such cases, please note that the same rules apply:

- The individual I-FDP implementation should be strongly related to the group I-FDP implementation, and not something different / unrelated. A project that focuses on one main theme with deeper integration would have higher value, than a project that has many smaller unrelated components and lower integration complexity.

- The same criteria will be used to judge your chosen I-FDP, namely:
  Functionality, **Robustness and reliability**, **User Interface** and **User Experience**

OTHER OPTIONAL DEVICES:

The OLED screen is a compulsory device for the I-FDP. All other devices are optional, and they are not needed for a project to be considered excellent.

Usage of devices (except computer keyboards and mouse) not provided by the lab are not allowed.
Hardware is not assessed in EE2026, and no additional marks are given for optional devices and the usage of their corresponding design sources.

The lab has the devices listed below and you can only request to sign out for these devices **during EE2026 lab sessions**. The lab counter does not distribute EE2026 components outside EE2026 lab sessions hours. **These external devices will not be explained, and design sources will not be provided. No additional marks are given for their usage**:

- PMOD for audio input
- PMOD for audio output
- VGA output to HDMI converter

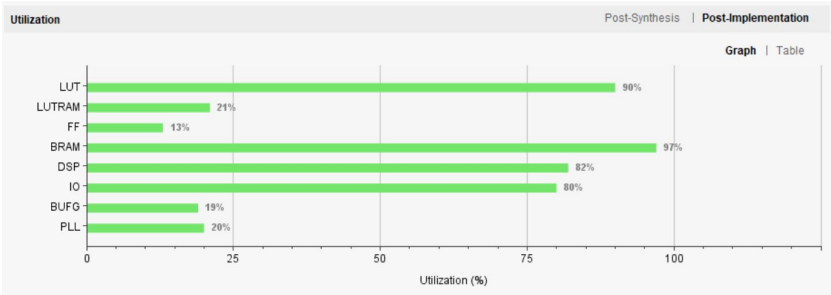
If you need to connect a Basys3 development board to another Basys3 development board, please seek **explicit email approval** from your lab assistant **before** making the connection. Connections between multiple Basys3 boards require you to **provide a schematic of which ports will be connected via wires** in the approval request email. **Do not request for Basys3 boards connections without providing the exact port connections schematics**. Also, wires will need to be connected to appropriate headers. Do not use wires without headers.

FPGA RESOURCE LIMITATIONS

All teams have the same amount of FPGA resources available and cannot exceed the resource limits of the FPGA. If you are not re-using parts from the basic tasks in I-FDP, consider removing them to release some FPGA resources.

Team members need to plan carefully, limit the number of components / capabilities or implement them more efficiently, and **INTEGRATE them EARLY and regularly**, to monitor their FPGA resource usage which might increase exponentially during integration.

The FPGA resource utilisation can be found in the Project Summary -> Utilization -> Post-Implementation. For example:



6. I-FDP PROPOSAL DOCUMENT (UPLOAD TO CANVAS AT THE END OF WEEK 8)

A description of the group theme your group has chosen needs to be **submitted to CANVAS**. The description should be no more than 2 pages (1 sheet), and should make use of the template below:

I-FDP PROPOSAL DOCUMENT	
Group ID: SX_YZ	**Description** **(Pictures / Drawings / Sketches / Gantt Charts are optional and not necessary)**
Group Theme: *Title of the group theme*	*Describe the overall group theme that your team wants to implement on the Basys3. If you are choosing the default FPGA Based calculator, this would involve indicating the overall functions of your calculator. The description for this part is recommended to be between 200 to 400 words. Minimum of 100 words required.*
Group Member 1: *Name of group member*	***Member responsibility:*** *"Describe what this group member is required to do with respect to the group theme"* ***Week 8 tasks:*** *"Describe what the student has **already done** for this week"* ***Week 9 tasks:*** *"Describe what the student will be doing for this week"* ***Week 10 tasks:*** *"Describe what the student will be doing for this week"* ***Week 11 tasks:*** *"Describe what the student will be doing for this week"* ***Week 12 tasks:*** *"Describe what the student will be doing for this week"* *The description for this part is recommended to be between 200 to 400 words and can be in bullet points format. Minimum of 100 words required.*
Group Member 2: *Name of group member*	***Member responsibility:*** *"Describe what this group member is required to do with respect to the group theme"* ***Week 8 tasks:*** *"Describe what the student has **already done** for this week"* ***Week 9 tasks:*** *"Describe what the student will be doing for this week"* ***Week 10 tasks:*** *"Describe what the student will be doing for this week"* ***Week 11 tasks:*** *"Describe what the student will be doing for this week"* ***Week 12 tasks:*** *"Describe what the student will be doing for this week"* *The description for this part is recommended to be between 200 to 400 words and can be in bullet points format. Minimum of 100 words required.*
… More members	*… More members*
Group Integration:	***Week 10 integration:*** *"Describe what parts will be integrated for this week"* ***Week 11 integration:*** *"Describe what parts will be integrated for this week"* ***Week 12 integration:*** *"Describe what parts will be integrated for this week"* *The description for this part is recommended to be between 100 to 200 words and can be in bullet points format. Minimum of 50 words required.*

Your team can still make changes to the actual implementations in week 8, 9, 10, 11 and 12.
There is no need to update the above document after submission.

UPLOAD DETAILS FOR WEEK 8:

- One **single PDF** submission **per group**
- File name format: **Week_8_SX_YZ**, where **SX_YZ** is your group ID
  (See PDF on CANVAS near submission time).
- Upload in the CANVAS folder named: **Week 8 I-FDP Proposal Submission**
- Deadline for submission: **Saturday 14th March 2026, 10 PM**

7. FINAL I-FDP SUBMISSION INSTRUCTIONS – ASSESSED IN WEEK 13

On average, the demonstration of the I-FDP in week 13 should have sufficient components for a duration of 2 to 5 minutes for the team, excluding questions asked by the assessor. Questions by the grader can extend the duration to more than 10 minutes.

Ensure that your project demonstration does not require the assessor to wait for very long, or to go through a lot of difficult steps, to see specific components / capabilities of your I-FDP. The assessor will skip those components / capabilities if there are time limitations, and this will affect the marks given.

There are two items (ITEM A, and ITEM B) to submit to CANVAS, and ITEM C to complete on CANVAS:

Official Lab Session	ITEM A (Submit on CANVAS) Project Archive	ITEM B (Submit on CANVAS) Report	ITEM C (Complete on CANVAS) Peer Feedback
Monday	**Deadline:** Friday 10th April 2026, 10:00 P.M.	**Deadline:** Sunday 12th April 2026, 10:00 P.M.	**Deadline:** Thursday, 16th April 2026, 10:00 P.M.
Tuesday			
Wednesday			
Thursday			
Friday			

Deadlines cannot be extended. Plan early on how to use the 6 weeks (Weeks 7 to 12) allocated for the project.

ITEM A: Project Archive Submission

- Only **ONE** Vivado project archive submission **PER team**. **The archive should not exceed 400 MB in size.** (Excess of 400 MB in size is only allowed with prior approval from the lab instructor)

- All functions must be present in **one single bitstream**. Ensure that your **one single bitstream** has been successfully generated and tested on your Basys 3 development board **BEFORE** archiving your Vivado workspace for CANVAS upload. Download your CANVAS archive after uploading. Unzip it and check if you can run your **one single bitstream** correctly.

- The **one single bitstream** must show only the I-FDP. The basic tasks that have already been assessed in week 8 are not required in the bitstream. If you are not re-using parts from the basic tasks, they are best removed to release more memory on the Basys3 board.

ITEM B: Report Submission in PDF Format:

Grading of the I-FDP is based on the description given in the report, with the described components / capabilities validated through the project demonstration to the assessors in week 13. **Understand the requirements below while writing the report**:

Sheet A (Pages 1 and 2) – Component / capability description part:

- Use the template shown on the last page. Note of the placement of the group ID, name, and matriculation.

- Consists of these sections: One for the group theme, and one section for each team member.
  - 4 members team: Total amount of words for the **5 sections combined** must **not exceed 500 words**.
  - 3 members team: Total amount of words for the **4 sections combined** must **not exceed 400 words**.
  - 2 members team: Total amount of words for the **3 sections combined** must **not exceed 300 words**.

- Summarise your components / capabilities descriptions clearly and concisely.
  - The special / interesting capabilities can be in bold characters, but do not overuse bold characters. Less than 20% of words for the description are allowed to be in bold characters.
  - If optional devices (Eg. Computer mouse, computer keyboard) are used, they should be clearly indicated in bold.
  - Usage of bullet points or numbering are allowed.
  - Instructions on how these components / capabilities can be operated by the user are optional

- Showing components / capabilities during the project demonstration to the assessors, but not described in the report, will not be given marks.

- Claiming for components / capabilities in the report, but which has not been implemented and demonstrated in the one single bitstream for the team, negatively affects the individual marks and team marks.
  - The team is responsible in ensuring that each team member has accurately indicated the components / capabilities that is implemented in the one single bitstream.
  - Demonstrating in a separate bitstream, such as due to running out of FPGA resources after integration, is not accepted.

- Images that support your components / capabilities need to be clear and in colour
  - Real images from the OLED screen are required, instead of simulated ones from the computer
  - Pictures should show the parts being focused on. For example, do not show the whole Basys3 board if only the 7-segments values are the parts in focus.

Sheet B (Pages 3 and 4) – Course feedback and references part:

- This sheet of A4 paper must only consist of course feedback and references.

- Include **course related feedback** (Ideally, no feedback about teachers in the report. It must be about the course):
  - Feedback, whether positive or negative, **DO NOT** have any effects on your grades.
  - Negative feedback with constructive comments is encouraged in this report, as they will allow two-way communications between students / teachers during the project presentation, to understand what to improve.
  - Anonymous feedback of teachers, or course feedback that don't need two-way communications, can be given on the official NUS end-of-semester feedback.

- **Include references used.** (GitHub, Websites, YouTube, Chatbots, AI, etc…)
  - **Compulsory** to include references to **open-sources codes**. Full penalties apply if open-source codes are used without clear references.
  - Usage of **Artificial Intelligence (AI)** or **automated code generators** must be **clearly indicated and documented**. You can use more pages (Page 5, 6, 7, etc…) to document those usage if AI has been used.

- Full penalties apply for the project if **open-source codes / AI / code generators / chatbots** have been used (These are allowed to be used) but **not properly documented on how they have been used in the project**.

ITEM C: Peer Feedback

Details for the peer feedback will be released through CANVAS announcement in week 12.

TEMPLATE FOR SHEET A (PAGES 1 AND 2) OF THE REPORT:

Text in black are the text that needs to be present in your report according to your custom details.
Text in grey are only for explanation purposes here and is not needed in your report.

Team: **SX_YZ**	Name of group theme: **A-mazing graphs game…**	**55 words**
The description of the group theme is written in this section. *Count the number of words (Group theme) from this section.* *Describe the group theme, indicating what the system can do overall. By reading this description, the user should know what the whole system is capable of and what are the star components / capabilities of the system, if any. The group theme MUST be a superset of the individual components, and not something different*		*Take only one picture of the whole system, showing where the OLED (and other devices you have used) are connected. The system should be running with your bitstream loaded, and the OLED screen / LED / Segments are showing something*
Student A: **Alice (A0000000X)**	Name of component: **Arithmetic process of…**	**88 words**
The description for student A is written in this section. *Count the number of words (Student A) from this section.* *Example (Based on a past project):* • *Calculator with 2 decimal point precision* • *Q16.8 fixed point arithmetic* • *Avoiding overflow through widening to Q48.8…* • *Shift-add algorithm for LUT reduction…*		
Student B: **Betty (A0000000X)**	Name of component: **Physics simulation of …**	**65 words**
The description for student B is written in this section. *Count the number of words (Student B) from this section.* *Example (Based on another past project):* • *Verlet integration: Numerically stable…* • *Multipass iterative constraint resolver: Allows item of arbitrary length* • *Randomized wind using 16-bit LFSR…*		
Student C: **Charlie (A0000000X)**	Name of component: **Windows interaction with…**	**128 words**
… *Example (Based a separate past project):* *The number in bracket used here allows for image matching* • *Gesture recognition from mouse (1)* • *Adjustable cursor and size: For ease of viewing* • *Speed of scrolling based on pushbutton press duration (2)*		(1) (2)
Student D: **Dawn (A0000000X)**	Name of component: **Interactive graph for…**	**93 words**
… *Example (Based a different past project):* • *Implemented a viewport controller that maintains zoom exponent and offset and it also acts as a movable and scalable window into a world coordinate system.* • *Intuitive mouse wheel that performs a smooth geometric zoom about the centre of the screen... (1)*		(1)

Colour Requirements to Ensure Accurate Grading:

- Highlight **TEAM SECTION IN PASTEL (OR LIGHT) PINK** (Do not use dark / bright pink)
- Highlight **STUDENT A SECTION IN PASTEL (OR LIGHT) RED** (Do not use dark / bright red)
- Highlight **STUDENT B SECTION IN PASTEL (OR LIGHT) BLUE** (Do not use dark / bright blue)
- Highlight **STUDENT C SECTION IN PASTEL (OR LIGHT) YELLOW** (Do not use dark / bright yellow)
- Highlight **STUDENT D SECTION IN PASTEL (OR LIGHT) GREEN** (Do not use dark / bright green)

Bring **TWO printed copies** of your report in **COLOUR**, when you come for your assessment.